

KALAT, Pakistan

WMO: 416960

Lat: 29.033N

Lon: 66.583E

Elev: 6617

StdP: 11.51

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.8	21.1	-1.5	6.5	37.5	2.0	7.8	35.4	10.5	41.7	7.3	45.0	0.3	N/A	0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.5	92.7	61.5	91.0	60.5	89.3	59.3	68.1	83.7	65.0	80.7	63.1	80.2	1.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.9	114.3	74.9	59.9	98.9	74.3	56.7	87.8	74.7	36.9	83.6	34.0	80.7	32.4	80.4	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.5	5.5	4.6	DB	12.2	95.7	1.7	2.1	10.9	97.2	9.9	98.5	8.9	99.7	7.7	101.3
			WB	9.3	68.3	1.3	2.9	8.4	70.5	7.6	72.2	6.9	73.8	5.9	76.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.7	37.8	41.4	50.3	59.7	67.3	73.9	76.9	74.6	67.1	56.6	46.5	39.6	(6)
(7)		DBStd	14.71	5.45	7.20	6.05	6.26	5.22	4.22	2.98	3.27	4.71	5.16	5.28	5.23	(7)
(8)		HDD50	1160	379	249	70	5	0	0	0	0	0	5	129	323	(8)
(9)		HDD65	3812	844	660	456	178	37	1	0	0	32	263	554	787	(9)
(10)		CDD50	3985	0	8	79	297	536	718	833	764	513	211	25	1	(10)
(11)		CDD65	1163	0	0	0	20	108	269	368	299	95	4	0	0	(11)
(12)		CDH74	14833	0	0	16	336	1584	3368	4385	3406	1509	227	2	0	(12)
(13)		CDH80	5912	0	0	1	48	474	1457	2056	1411	429	36	0	0	(13)
(14)	Wind	WSAvg	1.0	1.0	1.3	1.0	0.9	1.2	0.9	0.9	0.7	1.0	0.9	0.8	(14)	
(15)	Precipitation	PrecAvg	3.8	1.1	0.9	0.9	0.3	0.1	0.2	0.5	0.2	0.0	0.2	0.6	(15)	
(16)		PrecMax	12.1	5.1	5.3	5.0	1.8	1.5	2.8	3.0	2.4	1.4	1.0	2.8	4.8	(16)
(17)		PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	3.5	1.5	1.2	1.5	0.5	0.3	0.6	0.7	0.5	0.2	0.2	0.6	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.9	68.1	76.4	84.8	89.8	93.8	94.4	94.5	93.1	87.6	71.5	65.2	(19)
(20)			MCWB	42.0	47.1	50.5	54.5	57.1	58.3	60.7	65.9	73.9	64.5	47.3	44.3	(20)
(21)		2%	DB	57.8	63.3	72.2	80.7	87.6	91.8	92.7	91.4	87.2	80.1	68.3	60.0	(21)
(22)			MCWB	41.1	45.0	49.8	54.6	56.4	57.9	60.3	63.7	61.1	52.6	46.7	42.6	(22)
(23)		5%	DB	54.7	59.9	68.6	77.8	85.2	89.9	91.1	89.2	84.6	76.5	65.0	57.4	(23)
(24)			MCWB	39.9	43.9	48.0	54.1	55.7	57.8	60.0	60.1	57.6	51.8	45.4	41.2	(24)
(25)		10%	DB	51.4	56.4	64.9	74.8	82.7	87.7	89.4	87.3	82.4	72.7	61.8	54.1	(25)
(26)			MCWB	38.5	42.0	46.6	53.1	54.9	57.4	59.5	58.7	55.9	50.1	44.3	39.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.6	49.7	54.8	63.9	62.6	66.5	67.6	78.2	76.7	65.6	51.8	47.9	(27)
(28)			MDCB	53.4	63.8	68.1	76.0	82.8	77.4	83.0	90.9	89.0	83.8	64.2	56.4	(28)
(29)		2%	WB	43.3	46.9	51.5	59.2	59.2	64.2	65.4	67.5	69.4	55.7	49.2	44.6	(29)
(30)			MDCB	53.2	59.5	67.3	72.3	81.9	80.1	82.6	84.7	83.4	74.2	62.7	55.5	(30)
(31)		5%	WB	41.7	44.9	49.5	55.7	57.2	62.6	63.9	64.7	61.0	52.8	47.1	42.6	(31)
(32)			MDCB	51.9	57.6	65.5	72.7	80.9	80.1	82.1	79.5	75.4	72.2	62.2	54.5	(32)
(33)		10%	WB	40.0	42.8	47.6	53.8	55.9	60.3	62.4	62.5	57.8	50.9	45.2	40.9	(33)
(34)			MDCB	50.3	55.0	62.5	72.9	79.8	81.1	81.8	79.0	78.4	70.6	60.2	52.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	25.1	23.2	24.3	25.3	27.6	27.3	25.5	26.1	30.1	30.8	28.4	26.7	(35)
(36)			MCDBR	29.8	28.6	30.5	28.1	30.8	29.0	27.0	28.1	31.9	34.1	31.5	31.1	(36)
(37)			MCWBR	18.0	15.3	14.5	13.2	11.1	10.5	9.9	10.9	14.1	14.5	15.7	16.9	(37)
(38)		5% WB	MCDBR	25.9	24.6	26.7	25.6	27.8	25.1	24.7	23.4	28.6	30.8	27.8	26.3	(38)
(39)			MCWBR	18.4	15.9	16.4	16.4	12.4	12.6	11.6	13.6	18.4	15.1	15.7	16.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.288	0.334	0.372	0.425	0.433	0.466	0.572	0.506	0.385	0.326	0.309	0.278	(40)	
(41)		taud	2.401	2.264	2.126	1.996	1.975	1.898	1.696	1.832	2.123	2.263	2.322	2.431	(41)	
(42)		Ebn at Noon	303	295	290	277	273	263	236	251	281	294	292	302	(42)	
(43)		Edh at Noon	31	39	48	57	58	63	76	66	48	39	34	29	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1226	1443	1864	2151	2387	2468	2238	2154	2081	1820	1426	1208	(44)	
(45)		RadStd	83	126	113	117	95	102	121	116	74	48	62	67	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page